UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	OCTOBER 2026
TIME	2 HOURS

NSTRUCTIONS TO CANDIDATES

1. This question paper consists of 5 questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) a one – page Appendix 1
   iii) an Answer Booklet - provided by the Faculty
   iv) a graph paper - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 7 printed pages

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QUESTION 1

a) Solve the inequality $2x^2 - 4x - 16 \leq 0$.

(5 marks)



Figure: critical values.

b) Solve the inequality $|-4x - 2| \geq 8$. Express the answer in interval notation and on a number line.

(5 marks)

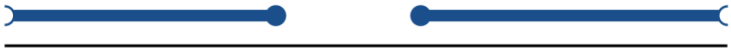


Figure: solution on a number line.

c) Sketch $y = 2|x + 2| - 2$. State the vertex, the y-intercept and any x-intercepts.

(5 marks)

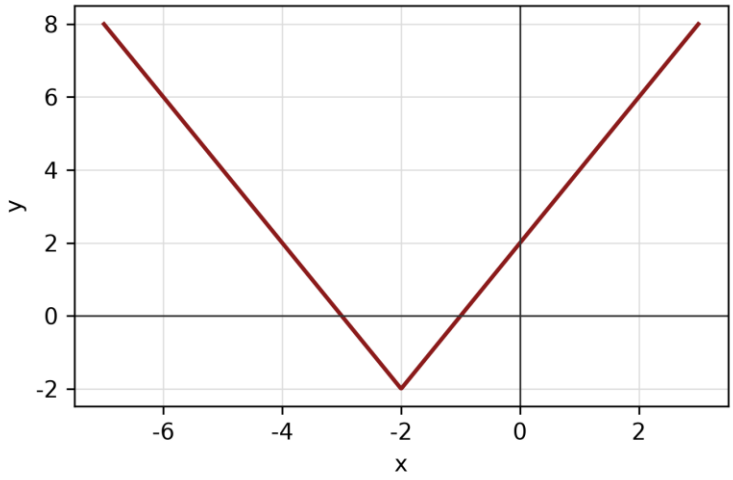


Figure: absolute-value graph.

QUESTION 2

a) The piecewise function is defined by
$f(x) = 3 - x$, if $x < -1$;
$f(x) = -2$, if $-1 \le x \le 2$;
$f(x) = x^2 - 1$, if $x > 2$.
Sketch the graph and find f(-2), f(0) and f(3).

(5 marks)

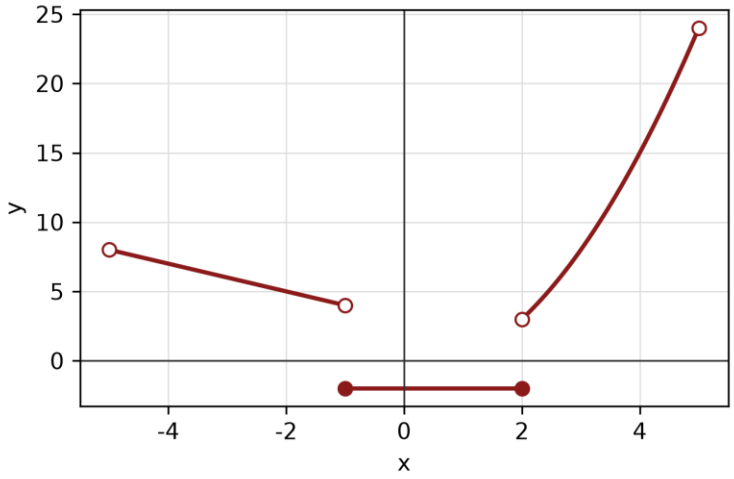


Figure: piecewise graph.

b) Sketch $y = |x - 2| + 1$ for $-2 \le x \le 6$. State the domain and range.

(4 marks)

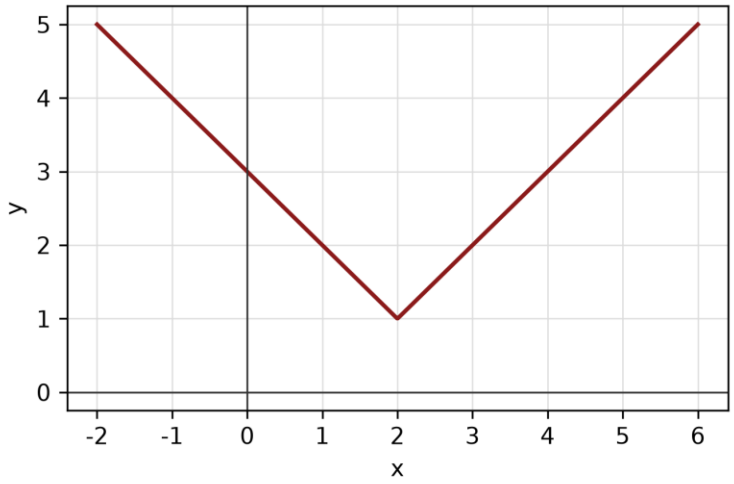


Figure: function graph.

c) The endpoints of a diameter of a circle are (4, 4) and (7, 4). Find the centre, radius and equation of the circle.

(5 marks)

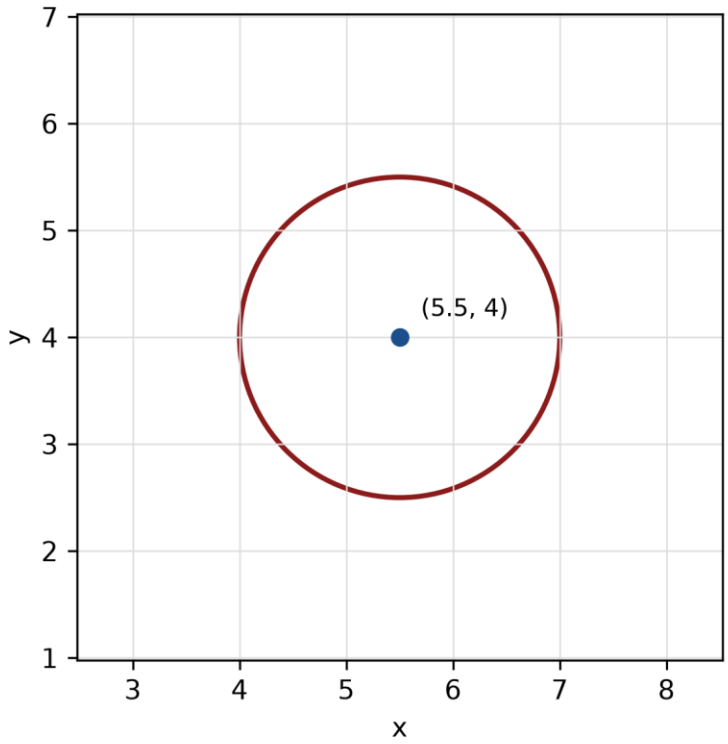


Figure: circle from diameter.

QUESTION 3

a) Given $z_1 = 4 - 4i$ and $z_2 = 2 - 5i$, express z_1/z_2 in the form a + bi.

(5 marks)

b) Evaluate i^20 and simplify completely.

(3 marks)

c) Show that x - 3 is a factor of f(x) = 3x^3 - 5x² - 17x + 15 by using the factor theorem. Hence factorise f(x) completely.

(5 marks)

QUESTION 4

a) Write a system of inequalities that describes the shaded triangular region with vertices (0, 0), (7, 0) and (0, 6).

(5 marks)

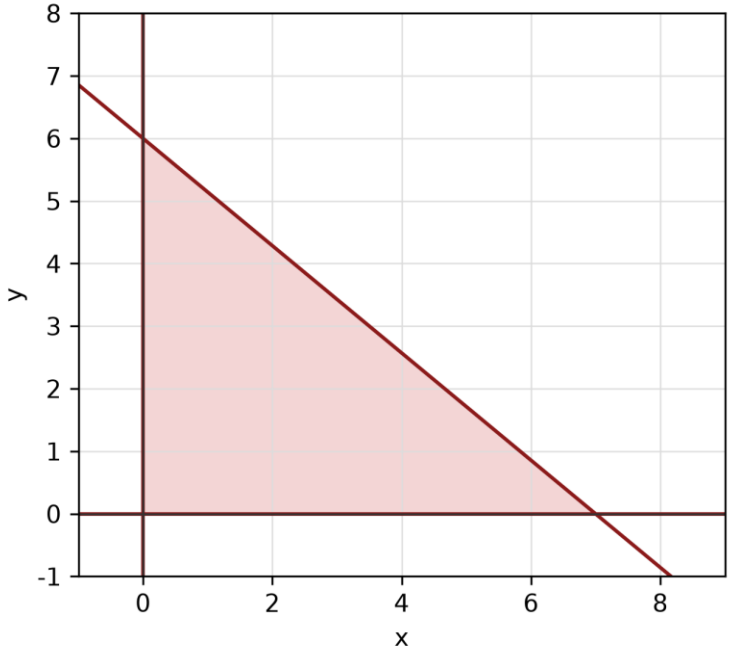


Figure: shaded triangular region.

b) Solve the simultaneous equations
-4x + y = 0
-x - 2y = 9.

(4 marks)

c) Find the standard equation of the parabola with focus (2, 7) and directrix y = 3. State its vertex.

(5 marks)

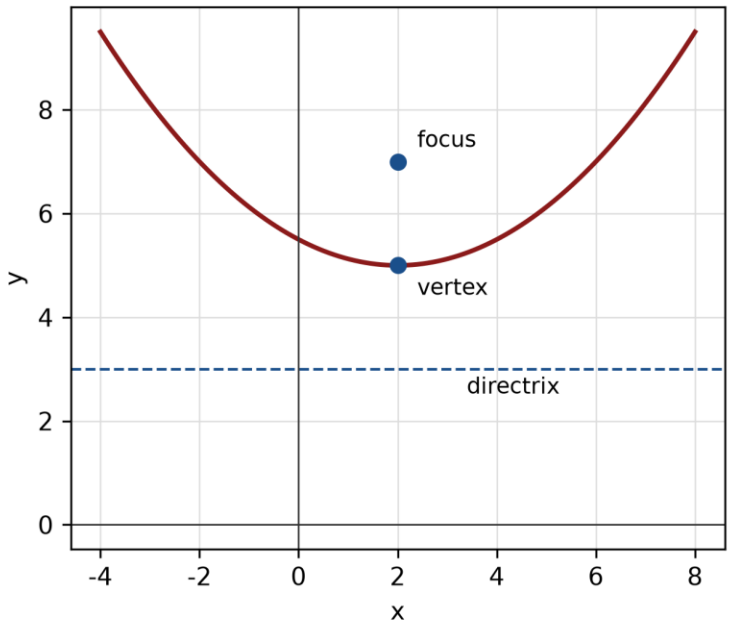


Figure: parabola with focus and directrix.

QUESTION 5

a) Given sin A = 8/17 and A is in quadrant 1, find the exact values of cos A, tan A and sec A.

(4 marks)

b) A circle has radius 8 cm. Find the length of the arc subtended by a central angle of 180°.

(4 marks)

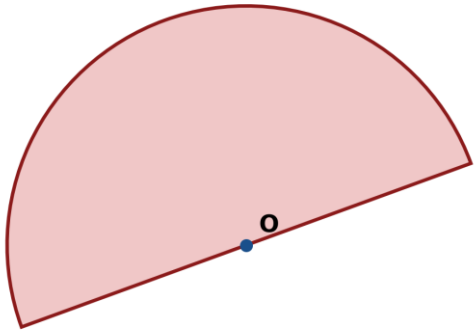


Figure: arc and sector.

c) Find the area of triangle ABC with a = 7 cm, b = 9 cm and included angle C = 90°.

(4 marks)

END OF QUESTION PAPER

APPENDIX 1

Formula sheet (as supplied with the paper).

$\sin^2 A + \cos^2 A = 1$

$\sin(A + B) = \sin A \cos B + \cos A \sin B$

$\cos(A + B) = \cos A \cos B - \sin A \sin B$

$\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$

$\cos 2A = \cos^2 A - \sin^2 A$

$\cos 2A = 1 - 2\sin^2 A = 2\cos^2 A - 1$

$\sin 2A = 2 \sin A \cos A$

$\tan^2 A + 1 = \sec^2 A$

$\tan 2A = 2 \tan A / (1 - \tan^2 A)$

Arc length, $s = r\theta$

Area of sector = $\frac{1}{2}r^2\theta$

$a / \sin A = b / \sin B = c / \sin C$

$a^2 = b^2 + c^2 - 2bc \cos A$

Area of triangle = $\frac{1}{2}ab \sin C$

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